Accessibility

Today's Topics

- Kinds of impairments
- Assistive technology
- Accessibility guidelines

Diversity of Ability

- Visual impairments
  - Color perception
  - Acuity ("legal blindness") inability to focus clearly
  - Total blindness
- Hearing impairments
  - Often varies with frequency
- Motor disabilities
  - Tremor and spasms
  - Muscle weakness and fatigue
  - Paralysis

Impairments Affect Everybody

- Aging
  - Reduced visual acuity
  - Hearing loss
  - Arthritis
- Overexposure
  - Noise-induced hearing loss
  - RSI Repetitive stress injury: motor impairment by excessive computer use
- Situational disabilities
  - Driving a car
  - Walking down the street
  - In a noisy environment

fat, textured handles (like the vegetable peeler)

Universal Design

- **Equitable use**
- Flexibility in use
- Simple & intuitive
- Perceptible information
- Tolerance for error
- Low physical effort
- Size and space for approach and use

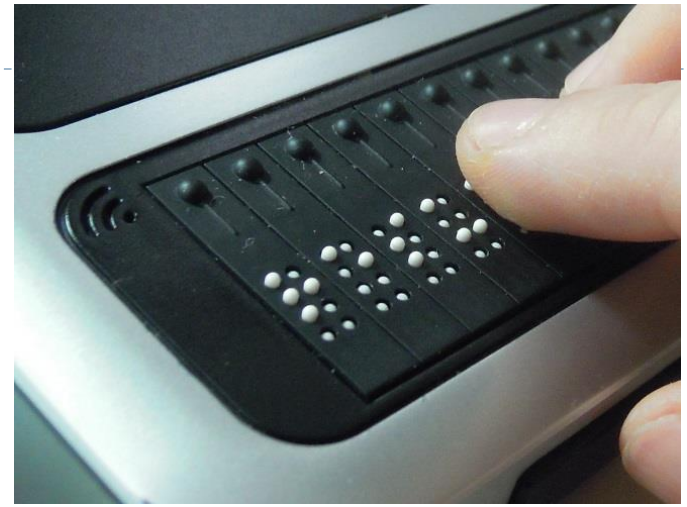
Image courtesy of enabledbydesign on Flickr.

Image courtesy of Jerry Kurjian on Flickr.

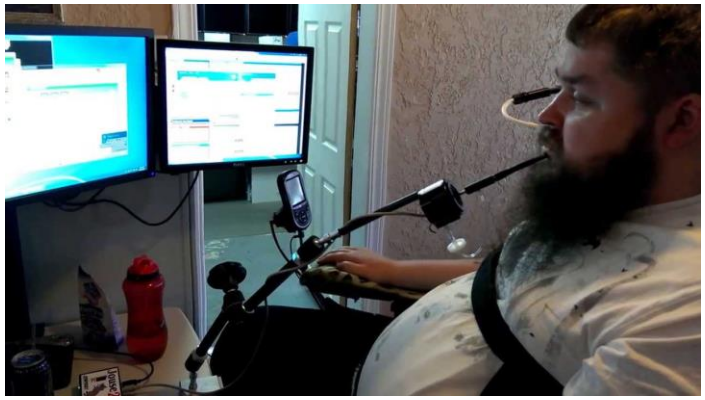
a sidewalk curb cut not only enables wheelchair users, but also parents with strollers, travellers with luggage

Assistive Technology

- Output
  - Screen magnifier
  - Screen reader
  - Braille display
  - Screen flashing on sound
- Pointing
  - Eye or head tracker
  - Puff-and-sip
  - Mouse keys
- Typing
  - Onscreen keyboards
  - Sticky keys
  - Speech recognition



http://www.districtdispatch.org/2016/07/good-news-braille-display-devices/



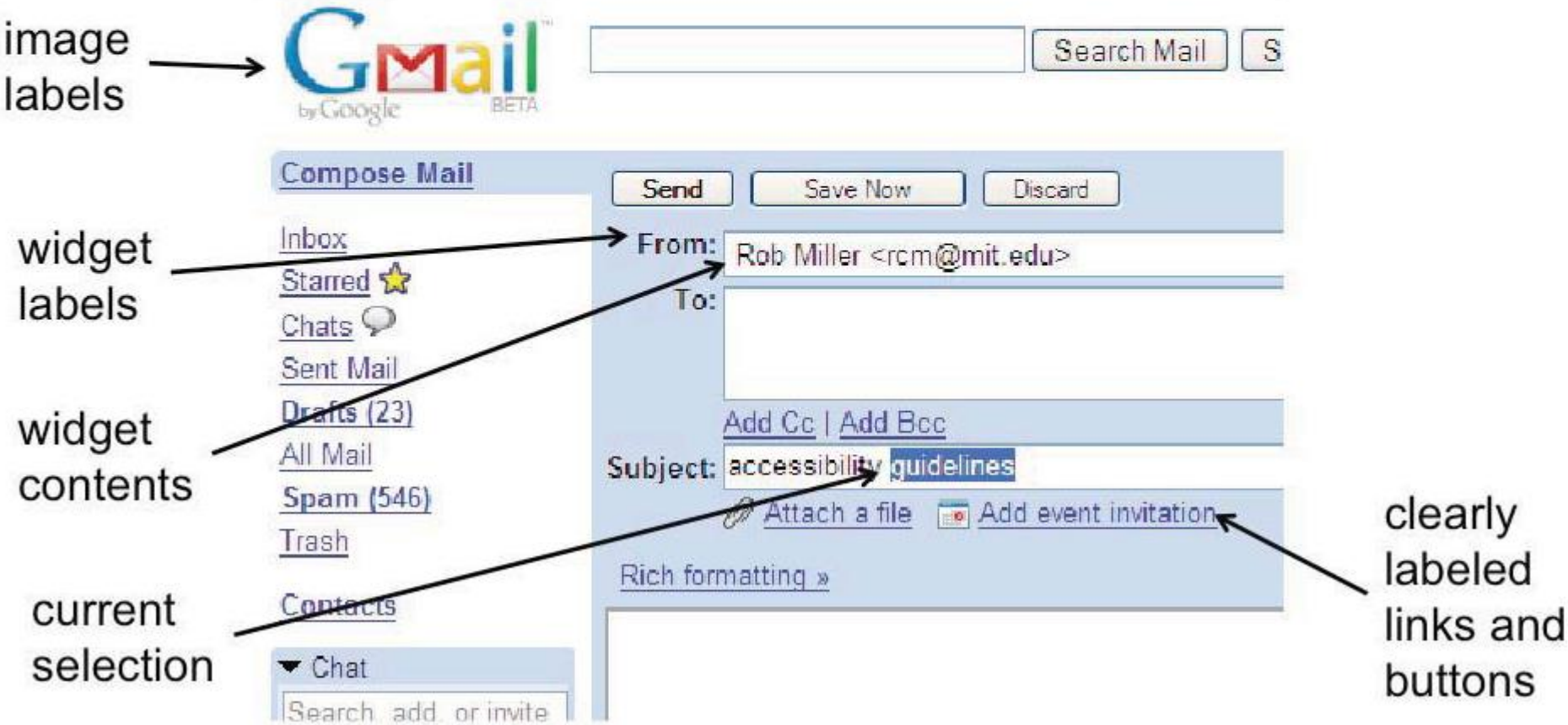
https://www.youtube.com/watch?v=fFi5_ctNFI0

Support Keyboard Access

- Pointing interactions should have keyboard alternatives
  - Menus should be controllable by the keyboard
  - Forms and links should be navigable by keyboard
  - Needed by motor-impaired and vision-impaired

Provide Text to Screen Readers

- Screen readers need to automatically transform the display into spoken language



Don’t Rely on Sound Alone

- Flash as well as beep
- Closed captioning for videos

User Control Over Colors and Fonts

- Allow user to choose high-contrast colors
- Allow user to enlarge fonts
- Don’t rely on color alone

Summary

- Accessibility is a universal problem
- Support alternative input (e.g. keyboards)
- Support alternative output (e.g. screen readers)